

CITYTWIN · REQUIREMENTS

CityTwin requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from CTW-CAL-001; R12 and R17 redefined and R20 added per CTW-DDR-001; R1 restated for the pull paths (D11); reference rates from the sibling TRL 3 notes
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

CityTwin requirements

The decisions in CTW-DDR-001 and CTW-DDR-002 (decided by Amish, 2026-09-25: go with recommendation) set these targets. R12 covers the pilot on a sheltered site at 0 to 35 °C air (D3, narrowed from 0 to 40 °C by DDR-002), the outdoor range is R20, R17 is a reported figure because the gateway is costed in TwinKit (D1), R10 is restated for the e-paper street kiosk, and R16 is set against the new `budget_usd` of \$750. The status column comes from the TRL 3 calculation note CTW-CAL-001 v0.2. Two requirements are **not met**: R18 (backup in the worst case, until TwinKit fits a larger pack) and R20 (street climate). R1 is **at risk**.

Table 1. Requirements.

ID	Requirement	Target	Status (CTW-CAL-001)	Verification (TRL 4 or later)
R1	Ingest every node type in the smart city set	All 9: seven over LoRaWAN through TwinKit; PotholeLog segment files and DockHub hourly aggregates pulled by outbound HTTPS from the operators' servers (D11)	At risk: six LoRaWAN types met on paper; the pull paths are proposed but not agreed with PotholeLog and DockHub; CrossSafe has no LoRaWAN uplink defined	Interface agreement with each sibling repo
R2	Capacity on one TwinKit gateway	50 nodes or more and 10,000 records per day or more	Met on paper: 46 nodes send 5,616 uplinks a day with 0.61 % lost (collisions and downlink blanking); 10,000 a day lose 0.71 %	Packet capture on a live gateway
R3	Accept only counts and levels	Each node type has a payload schema; any field outside it is rejected and logged; no images or audio stored	Met by design	Schema review; fuzz test of the decoder
R4	Suppress small counts in public outputs	People and vehicle counts below 5 per hour (D6) published as "fewer than 5"	Met by design	Review of export code
R5	No movement traces in public outputs	PotholeLog published per road segment only, no vehicle tracks or times finer than one day; DockHub per dock per hour, no card IDs	Met by design	Review of export code
R6	Open data export	Hourly CSV and GeoJSON with a documented schema, under CC BY 4.0 (D7), pushed to a public host	Met by design: 7,128 rows, 0.86 MB of CSV a day	Sample export checked against the schema
R7	One map for all node types	One layer per node type, node health, and an "as of" timestamp on every view	Met by design	Review of dashboard mock-up
R8	Data freshness	Map and kiosk show data no older than 30 min in normal operation	Met on paper, thin margin: 25.9 min worst on the kiosk, 15.6 min on the web map	Timing log on a live system
R9	Kiosk readable day and night	Readable in direct sun and at night at 1.5 m; key text cap height 7 mm or more	Met on paper: 41 px and 16.0 arcmin per capital; about 219 lx from the hood light dimmed to 0.3 W (1,459 lx at its full 2 W)	Mock-up viewed in sun and at night
R10	Kiosk responds to a button (restated, CTW-DDR-002)	Street kiosk: press acknowledged within 0.5 s by the button's LED ring, which stays lit until the new image is drawn; selected layer shown within 25 s. Indoor LCD variant: selected layer within 5 s (the TRL 2 target)	Met on paper: LED ring in about 30 ms; layer in 19.8 s. The indoor LCD variant is not modeled at TRL 3	Timed on a bench build
R11	Accessible kiosk	Operable parts 380 to 1,220 mm above ground (ADA forward reach); overhang from the post no more than 305 mm between 685 and 2,030 mm (protruding objects); same content on	Met on paper: buttons at 1,120 mm; overhang 200 mm (hood), 186.5 mm (cabinet with sun shield); web page not verifiable at TRL 3	Accessibility audit of the web page and a site review

ID	Requirement	Target	Status (CTW-CAL-001)	Verification (TRL 4 or later)
		a web page that meets WCAG 2.1 AA, linked by QR code and short URL		
R12	Pilot kiosk survives its climate (redefined, D3; range narrowed by CTW-DDR-002)	Sheltered or indoor site with no direct sun on the window; operate at 0 to 35 °C air; head IP54, cabinet IP55; panel within its 0 to 40 °C rating and the TwinKit pack within its 0 to 45 °C charge window	Met on paper: at 35 °C air the panel is at 35.1 °C (4.9 K margin) and the pack at 39.4 °C (5.6 K margin)	Temperature log in a climate chamber
R13	Kiosk stands up to wind	35 m/s gust with a factor of 1.5 without yield of post or anchors	Met on paper: post 26.9 MPa, 1.97 kN per anchor, with the sun shield	Structural design to the local code
R14	Electrical safety	Mains only inside the locked cabinet, behind a 30 mA RCBO and a surge protector; only 12 V SELV in the head; all metal earthed	Met by design; 12 V peak 26.7 W on a 60 W supply	Inspection by a qualified electrician
R15	Low running power	15 W average or less including the TwinKit gateway	Met on paper: 9.2 W (81 kWh a year), with the hood light dimmed to 0.3 W	Energy meter on a bench build
R16	Kiosk parts cost	Pilot kiosk (items 1 to 12, 14, 15) within <code>budget_usd</code> of \$750 (raised from \$600, CTW-DDR-002); the outdoor sun shield (item 19) is reported separately	Met on paper: \$739.00 (\$11.00 under); \$769.00 with the outdoor sun shield, \$19.00 over	Priced BOM
R17	Cost with gateway (redefined, D1)	Kiosk plus TwinKit gateway reported; the gateway is costed and budgeted in the TwinKit repo	Met on paper (reported): \$1,029.00	Priced BOM
R18	Honest in an outage	Screen keeps the last image with its timestamp when power fails; gateway rides through 2 h on its backup pack	Not met in the worst case: 2.40 h with a new pack, 1.63 h aged at 0 °C (TWK-CAL-001); a pack of at least 1.84 Ah (now 1.5 Ah) would meet 2 h and has been requested from TwinKit (CTW-DDR-002); the e-paper keeps its image by design	Outage test
R19	Secure by default	No inbound connections from the internet to the gateway; open data pushed and sibling data pulled by outbound connections only; no exposed ports or USB on the kiosk	Met by design	Security review
R20	Street kiosk climate (new, target after the pilot, D3)	Operate from -20 °C to +45 °C air in full sun	Not met: panel about 79 °C in low sun and 45.1 °C shaded at 45 °C air; a heated, insulated panel bay needs 13.5 W at -20 °C (fitted only where the partner city has frost); behind the sun shield the pack stays under 45	Chamber and field tests

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			°C in full sun up to 37.8 °C air, and is at 52.2 °C at 45 °C air	

■ Assumptions

- Node types and reporting rates follow the sibling TRL 3 calculation notes: CurbCount, HeatMap Node, FloodGauge and NoiseMap 96 uplinks a day; AirStreet 288; LoadZone 124 (100 state changes and 24 heartbeats). CrossSafe's hourly activation summary is a CityTwin assumption; CrossSafe has not defined a LoRaWAN uplink.
- Radio loss is calculated (CTW-CAL-001 section A) rather than assumed; the TRL 2 figure of about 3 % was a placeholder.
- The pilot kiosk (R12) stands on a sheltered site; the street range (R20) is the target after the pilot. Any outdoor site uses the cabinet sun shield and a shaded screen orientation as site rules (CTW-DDR-002).
- 35 m/s design gust is an estimate for a first check; the local wind code governs any real installation.